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Visual saliency decoding algorithm based on EEG signals☆

Li Li [a,b,c,*], Tongchao Lu [a,b,c]

[a] Beijing Laboratory of Advanced Information Networks, Beijing, China
[b] Beijing Key Laboratory of Network System Architecture and Convergence, Beijing, China
[c] Beijing University of Posts and Telecommunications, Beijing, China

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ABSTRACT

As a fundamental characteristic of the visual system, visual saliency has become a significant focus of research in recent years. However, existing methods rely solely on single-modality EEG signals with low signal-to-noise ratios, which significantly limits decoding accuracy and exhibits poor algorithmic interpretability. This study proposes an improved VSM generation method based on a two-dimensional Gaussian distribution to optimize VSM decoding. We designed a novel deep learning model, termed Multi-Encoder Single-Decoder with Channel Attention (MESD-CA), which integrates multimodal data, including electroencephalogram (EEG) signals and visual images, for VSM decoding. The MESD-CA model employs a two-stage training strategy to disentangle category-specific information from visual saliency features in EEG signals and incorporates visual image data to enable efficient decoding. Experimental results demonstrate that the MESD-CA model achieves localization precision and effective region F1 scores of 73.8% and 68.6%, respectively, significantly outperforming the Generative Adversarial Network (GAN). In addition, analysis of channel attention weights and the construction of brain topography reveal Fp2, FC4, and PO7 as key regions associated with the task. Distinct brain patterns were also observed across different object categories and fixation positions. This study provides new insights into the functional relationship between brain activity and the visual system, offering a foundation for applications in brain-computer interface (BCI) technologies.

1. Introduction

The human visual system has the ability to selectively focus on regions of greater interest or relevance within a visual field, thereby effectively reducing redundant visual information and enhancing the brain's efficiency in processing visual stimuli [1]. The mechanisms underlying the brain's processing of regions of interest are typically categorized into two types [2]. The first is the bottom-up process, which primarily relies on low-level visual attributes of the target region, such as color, luminance, and orientation [3,4]. The second is the top-down process, a task-driven mechanism influenced by the subject's intentions and cognitive goals [5]. The visual-related regions primarily encompass the occipital and parietal lobes [6]. The frontal lobe also plays a critical role in the formation of visual attention [7,8]. Furthermore, the frontal and parietal lobes collaboratively regulate attention allocation through their synergistic interactions [9]. The Visual Saliency Map (VSM) concept was first introduced in [10], providing a topographical representation of saliency stimuli in visual scenes. In this study, the VSM represents the probabilistic distribution of regions within a visual scene that are likely to attract the observer's attention. Specifically, the VSM is a single-channel image in which each pixel value indicates the probability of attracting attention, ranging from 0 to 1. Typically, VSMs are generated by analyzing visual features such as color, texture, and edges, providing a 2D representation that highlights the salient regions within the scene [11].

The generation of VSM relies primarily on fundamental image features or prior knowledge. On the one hand, analyzing basic image features, such as texture, color, and contours, effectively differentiates between salient and non-salient regions. Ref. [12] combined brightness and color as low-level visual features to propose a VSM generation method based on local contrast. Ref. [13] introduced a structurally guided statistical method for unique textures to detect salient regions. Ref. [14] employed the log-spectral residual of the images to propose a general object detection method. On the other hand, prior knowledge-based VSM generation methods utilize machine learning to extract objects with high-level semantic features. For example, [15] developed a fixation prediction model based on Support Vector Machines (SVM), combining low, medium, and high-level features to generate VSM. Ref. [16] proposed an end-to-end convolutional neural network

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* Corresponding author.
E-mail addresses: lili66@bupt.edu.cn (L. Li), luch@bupt.edu.cn (T. Lu).

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VGGNet to identify salient regions, demonstrating its capability in extracting high-level semantic features. Ref. [17] implemented a deeply supervised recursive convolutional neural network (DSRCNN) for image saliency prediction. However, these image-based VSM generation methods are often computationally intensive, with complexity increasing as the variety and quantity of objects in the visual field grow [18]. Therefore, a more accurate and simplified VSM generation method is crucial for optimizing decoding tasks. Additionally, these methods are highly dependent on image features or task-driven approaches and are not influenced by subjective human preferences. Since human attention to objects in the visual field is inherently subjective and dynamic, incorporating biomedical signals to explore the relationship between brain activity and visual saliency is crucial. When the brain performs different tasks, it exhibits specific functional states, with brain activity signals serving as key physiological markers of these states [19]. Given that visual attention originates primarily in the cerebral cortex and is closely associated with cortical activity, EEG signals provide an effective means of measuring cortical electrical activity. In addition, EEG is widely applied in neuroscience research [20] and brain-computer interface (BCI) systems [21] due to its noninvasive nature and high temporal resolution [22].

In recent years, the decoding of VSM from EEG signals has emerged as a novel research direction. Ref. [23] employed similarity analysis methods to examine the spatial distribution of event-related potentials (ERP) and visual saliency, revealing that the brain can rapidly compute the locations of salient regions in natural scenes, which helps guide attention and eye movements. Ref. [24] used complex videos of human activities in daily life to enhance visual stimuli and applied regression models, including Multiple Linear Regression (MLR), Support Vector Regression (SVR), and Kernel Ridge Regression (KRR), to establish the relationship between the EEG features and VSM. This study provided preliminary evidence to predict the visual saliency distribution using EEG data. Ref. [25] integrated the P300-based BCI paradigm with a fuzzy color extractor to identify regions of interest for humans. Ref. [26] proposed a model to calculate the importance of salient points for visual fixation positions and assessed the relationship between the importance of salient points and the amplitude of the EEG signal. Ref. [27] mapped EEG signals and VSM to a continuous feature space using two Variational Autoencoders (VAE) and combined an EEG VAE encoder with a VSM VAE decoder to form a generator for GAN. After training the GAN, the study successfully decoded VSM from EEG signals. Current research primarily focuses on combining the positional and semantic information in VSM but lacks effective disentanglement analysis. Considering the rich information embedded in EEG signals, integrating them with other modalities can fully exploit the potential of EEG data, thus significantly improving the accuracy of VSM decoding. Furthermore, existing studies have not analyzed the importance of different EEG channels. Since the cerebral cortex is functionally partitioned into distinct regions that perform specialized perceptual tasks [6], certain key cortical areas are likely critical for VSM decoding. Analyzing and enhancing these regions could further improve the precision and robustness of VSM decoding.

To address the above issues, we propose a novel visual saliency decoding algorithm based on a Multi-Encoder Single-Decoder with Channel Attention (MESD-CA) framework. First, to circumvent the complexity and instability inherent in GAN network training, we adopt an encoder–decoder architecture, which is structurally simple and yields stable training outcomes. Second, to enhance the information content of the model's input data, we incorporate visual image data as a multimodal input, performing feature-level multimodal fusion with EEG signals. Third, to enhance the interpretability of the model structure, we implement a two-stage training strategy that enables differentiated training of distinct encoder branches. Finally, we introduce a channel attention mechanism to address the shortcomings of existing methods in evaluating the importance of different electrode positions. The primary contributions of this study are summarized as follows:

(1) We design a novel experimental paradigm of visual attention. Real objects are presented to participants to enhance visual stimuli, and their positions within the visual field are dynamically adjusted. This breaks the conventional paradigm of discrete visual focus and makes the experiments more realistic.

(2) We propose a VSM generation method based on a two-dimensional Gaussian distribution. This approach simplifies the data annotation process and reduces the complexity of VSM decoding tasks.

(3) We propose the MESD-CA deep learning algorithm. Based on a multi-encoder single-decoder architecture, MESD-CA adopts a two-stage training strategy to disentangle semantic information and saliency information from EEG signals. It effectively integrates visual image features for decoding, achieving state-of-the-art performance.

(4) We introduce the channel attention mechanism into the MESD-CA model, which further improves the accuracy of decoding VSM from EEG signals. Channel attention weights are used to construct brain topography, enabling the analysis of the neural mechanisms underlying visual saliency and the identification of critical brain regions.

The remainder of this paper is organized as follows: Section 2 presents the proposed experimental paradigm, dataset, model architecture, and training strategy; Section 3 describes the evaluation metrics and experimental results; and Section 4 concludes the study.

2. Materials and methods

2.1. Experimental paradigm

Existing experimental paradigms fix the positions of visual stimuli in a few discrete locations [28–30]. However, a person's visual saliency continually shifts within their field of view. To further explore the relationship between visual focus and brain mechanisms under more flexible and natural conditions, we designed a new paradigm of visual attention stimulation, as shown in Fig. 1. The left side of Fig. 1 presents the equipment used in our experiment, which includes an EEG signal acquisition device (NeuSen W system), a visual image acquisition camera, and four devices observed by the participants (projector, fan, power strip, and electric kettle). Unlike traditional experiments in which stimuli are displayed on a screen, this study places real devices directly in front of participants to enhance the authenticity of visual stimuli. In addition to collecting EEG signals during the experiment, a camera was mounted on the center of the participant's forehead to capture visual image data, ensuring that the observed device remained within the visual image frame. Furthermore, efforts were made to keep the participants' heads stable to prevent changes in the field of view from affecting subsequent analyses.

The right side of Fig. 1 illustrates the experimental paradigm, where each row represents a session. Before each session begins, the positions of the devices are rearranged. Participants must observe the four devices in a fixed sequence within each session to ensure data balance. The orange boxes in Fig. 1 indicate which device the participant is currently observing, but such visual indicators are not used in the actual experiment. The participants' start and end observation times for each session and trial are controlled by a keyboard, allowing them to adjust the experimental process according to their needs and thus avoid low data quality caused by rigid experimental paradigms. Between each trial, there are a few seconds of adjustment time, which is also controlled by the subjects. During each trial for the respective device, participants are instructed to maintain their visual focus on the corresponding device and to continuously verbalize the device's name mentally, enhancing their attention toward the device. During the experiment, we simultaneously collected EEG signals and visual image data to ensure temporal consistency between the two datasets, providing a reliable foundation for subsequent analyses. Prior to the commencement of the experiment, all participants provided informed consent. This experimental study received ethical approval (No. BUPT-P-2025002) from the Beijing University of Posts and Telecommunications Ethics Review Committee.

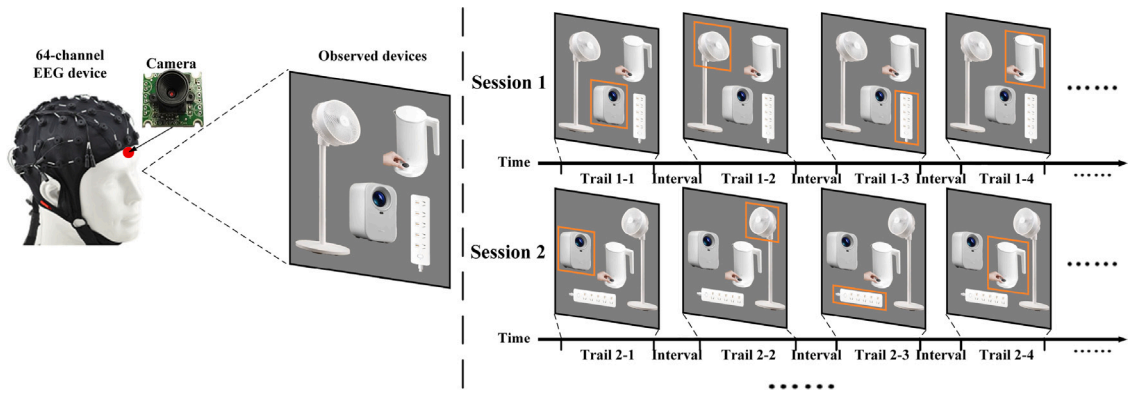

Fig. 1. (Left) Experimental equipment. (Right) Experimental paradigm.

2.2. Dataset description

EEG data were recorded using the Neuracle NeuSen W system with 64 electrodes, following the International 10-20 System [31] and sampled at 1 kHz. ECG electrodes were not included in the electrode configuration. Furthermore, four electrodes, HEOL, HEOR, VEOL, and VEOU, were used for electrooculography (EOG) measurements. As a result, 59 EEG electrodes were employed for the recordings. The recorded data were first divided into 80% for the training dataset and 20% for the test dataset. The EEG signals were segmented using a 1-second time window with 50% overlap, resulting in a data format of 59 × 1000 for each EEG segment. In total, 25190 samples were obtained for the training dataset and 8040 for the test dataset. The camera captured visual image data at a frequency of 10 Hz and synchronized with the EEG signals using timestamps. For each EEG segment, the visual image data corresponding to the first frame after the start of the time window were selected. Each data sample is associated with the category label of the object that the subject focuses on during the experiment.

2.3. Data preprocessing

EEG signals are microvolt-level electrical signals that are inevitably affected by noise and interference during data acquisition [32]. Previous studies have shown that the effective frequency range of EEG signals is distributed below 100 Hz [33]. Therefore, in this study, a Finite Impulse Response (FIR) bandpass filter was applied to EEG signals within 1 to 100 Hz to enhance the signal-to-noise ratio (SNR) [34]. A 50 Hz notch filter was also utilized to eliminate power line interference. The Independent Component Analysis (ICA) was then performed, referencing the EOG signals to remove ocular artifacts from the EEG data [35].

Visual image data were collected using a camera mounted at the center of the participant's forehead, ensuring the captured images aligned with the participant's field of view. All collected visual images were resized to 224 × 224 pixels, as shown in the first row of Fig. 2. The red dots in the images were manually annotated on the basis of the device category labels observed by the participants and were positioned in the center of the corresponding device. To address the uncertainties related to the shape, area and centroid of the effective region in current methods for generating a VSM, this study proposes a novel method for generating a VSM based on a 2D Gaussian distribution, with the annotated points in the image serving as the centers of the distribution. The specific calculation formula is as follows:

$$VSM = exp\left(-\frac{\|S-C\|^2}{2\left(\frac{max(h,w)}{\lambda}\right)^2}\right), \tag{1}$$

where C representing the coordinates of the central position, S representing the coordinates of any given point, λ represents the attenuation adjustment factor of the distribution, h being the height of the image and w being the width of the image.

After testing, an optimal λ value of four was determined. The final VSM example is shown in the second row of Fig. 2, where the red dot corresponds to the position of the red dot in the first row, representing the location of the observed object's center in the field of view. The regions surrounding the red dot exhibit higher values, indicating a greater likelihood of being observed, while distant areas have lower values, suggesting a lower probability of being noticed.

2.4. Model architecture

This study aims to explore the relationship between EEG signals and visual saliency using deep learning algorithms. Furthermore, considering that the category of the observed object is a significant factor influencing participants during the experiment and drawing on research related to object detection based on category in images [36], we hypothesize that the integration of category features from EEG signals with visual image data can assist in the decoding of the VSM. The overall structure of the proposed model is shown in Fig. 3(b). The model takes EEG signals and visual image data as input and outputs the predicted visual VSM. The VSM Generator illustrated below represents the VSM generation method based on a 2D Gaussian distribution, which is used to generate the ground-truth VSM. The model is trained by comparing the ground-truth VSM with the predicted VSM. The proposed model consists of three encoders, one decoder, and a channel attention module. It is referred to as the Multi-Encoder Single-Decoder model with Channel Attention (MESD-CA). The five main components of the MESD-CA model proposed in this study are detailed as follows:

2.4.1. Category encoder (CE)

The EEG signals collected in the experiment contain both the category information of the observed object and the location of the focus of the object in the visual field. To effectively decouple these two types of features from EEG signals, we first pre-train the CE through a classification task. As shown in Fig. 3(a), the EEG signals are fed into the CE, which consists of multiple layers of the Transformer Encoder and 1D CNN layers and outputs the category features. These features are then passed through a Classifier composed of 2D CNN layers to predict the category. Since there are four devices in this study, the task is a four-category classification, and the parameters of the CE and the Classifier are adjusted by calculating the Cross-Entropy loss between the predicted and true categories.

2.4.2. Channel attention module (CAM)

If we directly input the raw EEG signals into the feature extraction model, it would imply that all EEG channels contribute equally to the visual saliency decoding task, which is not the case. Numerous studies have shown that different regions of the cerebral cortex are

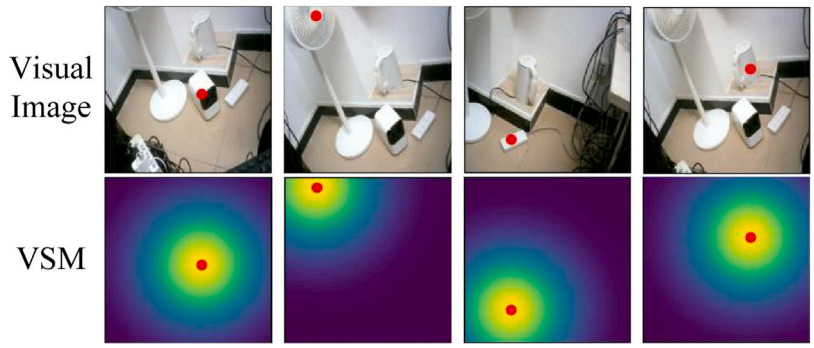

Fig. 2. (First row) Examples of visual image. (Second row) The corresponding VSM aligned vertically with the visual images.

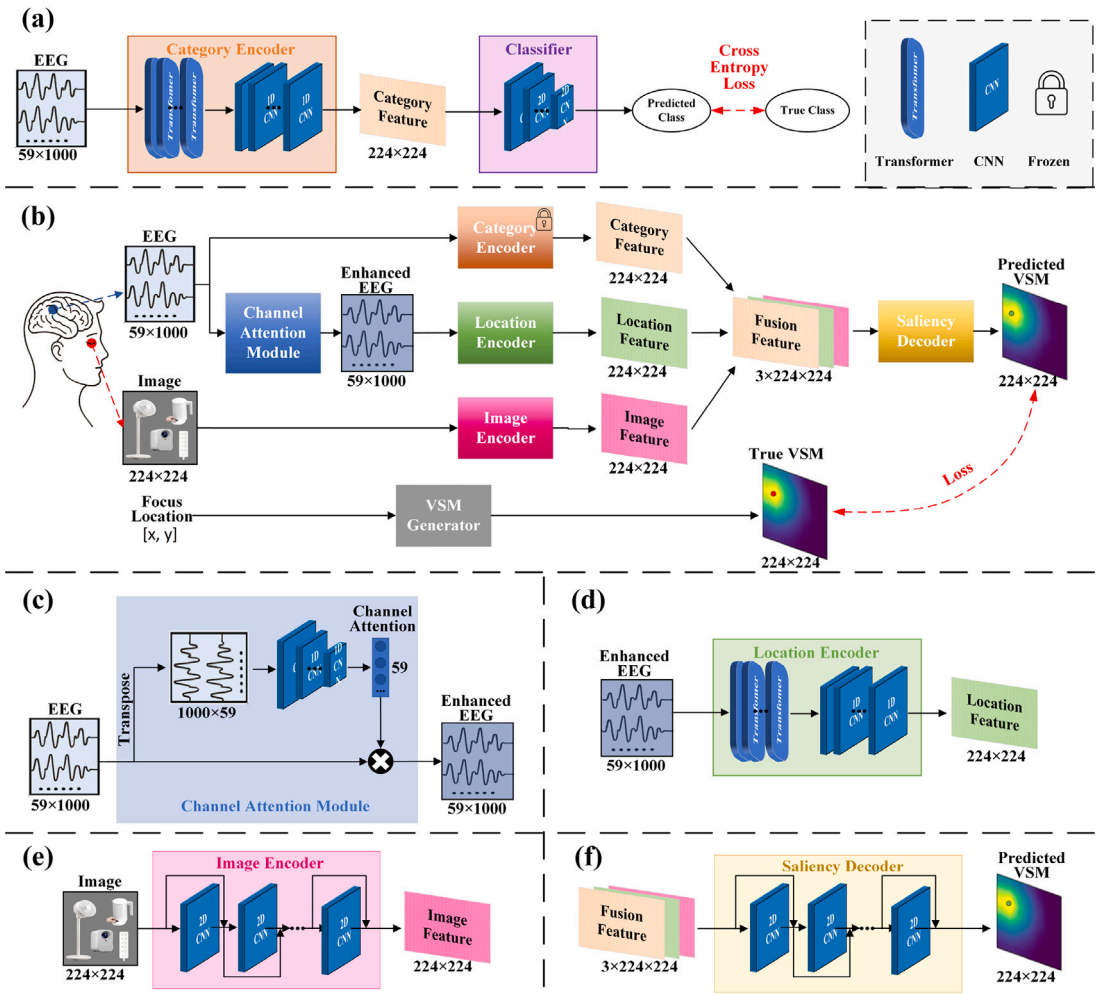

Fig. 3. (a) Step 1: Pretraining of the Category Encoder (CE) (b) 2: Overall architecture and training of the MESD-CA model. (c) Channel Attention Module (CAM). (d) Location Encoder (LE). (e) Image Encoder (IE). (f) Saliency Decoder (SD).

responsible for different functions [37]. For example, the early and higher visual cortices play different roles in decoding brain activity signals. Therefore, it is necessary to design a specialized mechanism to enhance the EEG signals from channels to be more relevant to visual saliency decoding while suppressing signals from irrelevant channels. To address this, we propose the CAM. As shown in Fig. 3(c), the input EEG signals are first transposed, treating the 59 channels at a given time as a lateral dimension. After passing through a series of 1D CNN layers, the model outputs attention weights for the 59 channels. The data from each channel are then multiplied by its corresponding attention weight, resulting in enhanced EEG signals.

2.4.3. Location encoder (LE)

During MESD-CA training, we keep the CE parameters fixed, preventing them from being updated. It ensures that the features extracted by the CE remain purely category-related and unchanged. In the current training task, which focuses on decoding visual saliency, the information on visual saliency in the EEG signals can only be extracted by the

LE. As shown in Fig. 3(d), the LE consists of Transformer Encoder layers and 1D CNN layers and is responsible for extracting location features from enhanced EEG signals.

2.4.4. Image encoder (IE)

Visual images contain rich information, including interrelated semantic and positional data of the devices. This module is designed to combine the category features extracted by the CE to assist in decoding the visual saliency. As shown in Fig. 3(e), the image encoder uses a 2D convolutional ResNet network to convert three-channel visual images to single-channel image features.

2.4.5. Saliency decoder (SD)

The CE, LE, and IE produce the category features, location features, and image features, respectively, each designed with a size of 224 × 224. These features are stacked as individual layers to form a fusion feature 3 × 224 × 224. This fusion feature is processed by the SD, where the architecture of the SD model is identical to that of IE and produces a predicted VSM. Subsequently, the loss value between the predicted VSM and the ground-truth VSM is computed to update the parameters of the entire MESD-CA model, excluding CE.

2.5. Training methodology

As mentioned earlier, the training of the entire model is divided into two stages. As shown in Fig. 3(a), during the first phase, the CE is pre-trained using a category classification task to ensure its ability to accurately extract category features from EEG signals, providing a reliable foundation for subsequent model training. The formula for calculating the Cross-Entropy loss between the predicted and true categories is:

$$\mathcal{L}_c = CrossEntropy(y, \hat{y}), \tag{2}$$

where $CrossEntropy()$ is the function used to calculate Cross-Entropy loss, y is the true label, and $\hat{y}$ is the predicted label.

In the second stage, we use the model shown in Fig. 3(b) to perform the visual focus decoding task. First, the CE parameters are fixed, and the LE, CAM, IE, and FD parameters are updated during the model training process. The total loss $\mathcal{L}$ is computed by calculating the pixel-wise loss $\mathcal{L}_{pw}$ between the predicted VSM and the true VSM, along with the deviation loss $\mathcal{L}_d$ between their distribution centers.

$$\mathcal{L} = \mathcal{L}_{pw} + \mathcal{L}_d = MSE(v, \hat{v}) + \alpha \cdot \|c - \hat{c}\|, \tag{3}$$

where v is the true VSM, $\hat{v}$ is the predicted VSM, $MSE()$ is the function that calculates the mean squared error between v and $\hat{v}$, c represents the position coordinates of the center of the ground-truth VSM distribution, while $\hat{c}$ denotes the position coordinates of the center of the predicted VSM distribution calculated based on $\hat{v}$. The formula for calculating $\hat{c}$ involves setting a threshold θ to determine the range of coordinates. First, we obtain the horizontal and vertical coordinate ranges where the sum exceeds θ, and then take the center of these ranges as the coordinates (p_h, p_v). p_h and p_v are the horizontal and vertical coordinates, respectively. The formula is expressed as follows:

$$p_h = \frac{\min(i \mid \sum_j \hat{v}_{i,j} > \theta) + \max(i \mid \sum_j \hat{v}_{i,j} > \theta)}{2}, \tag{4}$$

$$p_v = \frac{\min(j \mid \sum_i \hat{v}_{i,j} > \theta) + \max(j \mid \sum_i \hat{v}_{i,j} > \theta)}{2}, \tag{5}$$

where i and j are the column and row indices of the image, and $\hat{v}_{i,j}$ is the value of the $\hat{v}$ at pixel (i, j).

3. Results and discussion

In this section, we first introduce the comparison algorithm. Subsequently, we propose evaluation metrics for assessing model performance. Next, we visualize the VSM predicted by the model to evaluate its performance intuitively. Following this, we employ the evaluation metrics to compare differences between various algorithms and conduct ablation studies. Finally, we construct brain topographic maps using the channel attention weights output by CAM and analyze them.

To validate the effectiveness of MESD-CA, we selected a GAN-based VSM decoding algorithm as the comparison method [27]. Comprehensive ablation studies were conducted on data modalities and model architectures to verify the role of each component. The Baseline is the simplest VSM decoding model, comprising the LE and SD components. The MESD model is derived from the MESD-CA model by removing the CAM component. To ensure the reliability of the experimental results, all model architectures in this study were evaluated based on the average results of five independent repeated trials.

3.1. Evaluation metric

The model's predicted VSM output is intended to match the true VSM, which follows a 2D Gaussian distribution. However, the actual predicted VSM may deviate from this ideal state. Multiple metrics are needed to evaluate the model's performance comprehensively.

First, since the VSM values range from 0 to 1, we apply a threshold β to process the predicted VSM: The values below β are set to 0, and the values above or equal to β are set to 1, thus generating a predicted region (the area with a value of 1). If the true visual focus lies within this predicted region, it is considered correct; otherwise, it is incorrect. By tracking this, we can calculate the precision $P_l(\beta)$ of the algorithm's localization performance.

$$P_l(\beta) = \frac{n_c(\beta)}{N} \tag{6}$$

where $n_c(\beta)$ is the number of samples where the true visual focus falls within the predicted area after being filtered by threshold β, and N is the total number of samples.

Similarly, a threshold β can also be applied to the true VSM to obtain the true effective region $S_e(\beta)$. The overlap rate between the predicted region $S_p(\beta)$ and $S_e(\beta)$ can then be compared. We define two metrics: effective region recall $R_e(\beta)$ and effective region precision $P_e(\beta)$. $R_e(\beta)$ represents the proportion of the overlapping region $S_o(\beta)$ within $S_e(\beta)$, which evaluates the model's ability to detect the true effective region. If the area of $S_p(\beta)$ is sufficiently large, a better metric $R_e(\beta)$ can be obtained; however, this is not our objective. To obtain a suitable $S_p(\beta)$, we introduce $P_e(\beta)$. $P_e(\beta)$ represents the proportion of $S_o(\beta)$ within $S_p(\beta)$, which evaluates the accuracy of the predicted region of the model. Combining these two metrics, we derive a balanced measure, the effective region F1 score $F_e(\beta)$, with higher values indicating better model performance.

$$R_e(\beta) = \frac{S_o(\beta)}{S_e(\beta)}, \tag{7}$$

$$P_e(\beta) = \frac{S_o(\beta)}{S_p(\beta)}, \tag{8}$$

$$F_e(\beta) = \frac{2 \times R_e(\beta) \times P_e(\beta)}{R_e(\beta) + P_e(\beta)}. \tag{9}$$

3.2. Predicted VSM visualization

This study compares the predicted VSM outputs of the models with the ground-truth VSM and assesses the performance differences among various models. The final results are presented in Fig. 4. The leftmost column indicates different values of β, where $\beta = 0$ does not represent threshold filtering, with the original data output directly. The left side

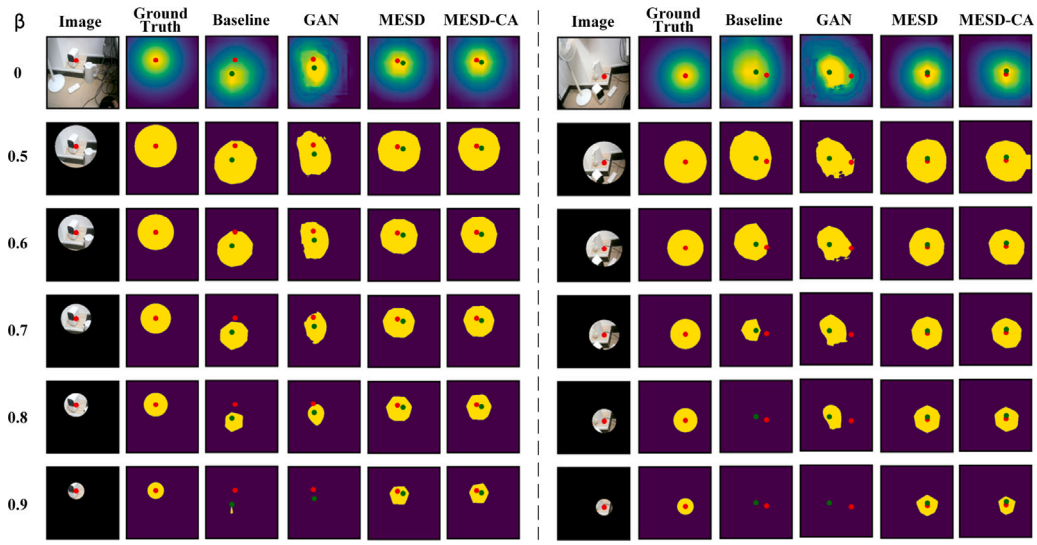

Fig. 4. Examples of visualized results for the original images (Image), true VSM (Ground Truth), and the predicted VSM outputs from various models (Baseline, GAN, VFDM1, VFDM2) at different values.

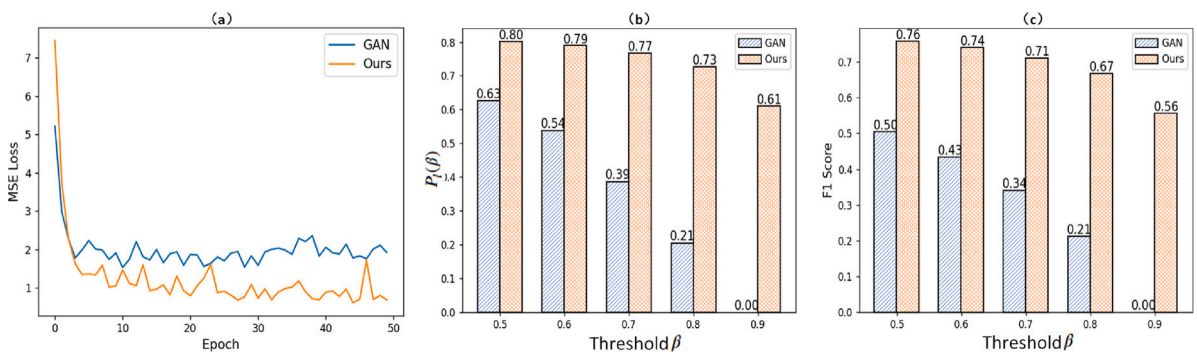

Fig. 5. Comparison of experimental results. (a) MSE variation curve of mean square error during training, (b) test results of $P_L(\beta)$ at different thresholds, and (c) test results of $F_E(\beta)$ at different thresholds.

presents an example of a projector, while the right side illustrates an example of an electric kettle. In each example, the leftmost column displays the masked image of the actual data after applying different threshold filters; the second column shows the binary valid regions of the ground-truth VSM after filtering with varying thresholds. The subsequent four columns correspond to the four algorithms. The red dots in the figure represent the true visual focus locations, while the green dots indicate the predicted visual focus locations.

In Fig. 4, by comparing the effective regions of VSM generated by MESD-CA and GAN models after threshold processing, as well as the positional relationships between the predicted centers and the actual object centers, we observed that the effective regions produced by MESD-CA are more closely aligned with the circular shape of the actual regions. Additionally, the predicted centers are more accurately aligned with the true centers, indicating that MESD-CA achieves superior VSM generation performance. Furthermore, while the Baseline model can predict VSM to some extent under low-threshold conditions, its prediction deviation is significantly greater than that of the MESD model. This demonstrates the effectiveness of our two-stage training strategy, which disentangles EEG information and integrates it with visual images for multimodal fusion.

3.3. Performance evaluation

Fig. 5(a) illustrates the MSE loss curves for the GAN algorithm and the proposed MESD-CA algorithm during the model training process. As shown in the figure, at the initial stage of training, the MSE of GAN is significantly lower than that of MESD-CA. This phenomenon is primarily attributed to GAN's pre-training phase, where two VAEs were employed for self-supervised training on EEG and VSM data, respectively. Consequently, the EEG VAE's encoder developed robust EEG signal feature extraction capabilities, while the VSM VAE's decoder exhibited strong VSM generation performance. The GAN generator, leveraging the strengths of both, enabled rapid adaptation to new tasks. However, as training progressed, the advantage of GAN's prior knowledge gradually diminished, whereas MESD-CA maintained strong learning capacity. In terms of converged loss values, the performance of the proposed MESD-CA is markedly superior to that of GAN.

To comprehensively evaluate and compare model performance, this study utilized test set data to assess the trained models, calculating evaluation metrics $P_l(\beta)$ and $F_e(\beta)$ based on their outputs. The results are presented in Fig. 5(b) and (c). In these figures, the horizontal axis represents the threshold β, which was assigned five values ranging from 0.5 to 0.9. Higher β values correspond to smaller effective regions, imposing stricter requirements on model performance.

As depicted in Fig. 5(b) and (c), MESD-CA consistently outperforms GAN across all metrics for all β values. Notably, at lower β values, the differences in metrics $P_l(\beta)$ and $F_e(\beta)$ between GAN and MESD-CA are relatively small. However, as β increases, the performance gap widens significantly. Specifically, MESD-CA's metric $P_l(\beta)$ decreases from 0.80 to 0.61, a reduction of 0.19, and metric $F_e(\beta)$ decreases from 0.76 to 0.56, a reduction of 0.20. In contrast, GAN's metric $P_l(\beta)$ drops

Table 1
Ablation experimental results.

Algorithm	Metric	Threshold β					Average
		0.5	0.6	0.7	0.8	0.9	
Image	$P_l(\beta)$	0.1108	0.0623	0.0322	0.0071	0.0000	0.0424
	$F_e(\beta)$	0.0976	0.0437	0.0106	0.0003	0.0000	0.0304
w/o IE	$P_l(\beta)$	0.3987	0.2247	0.1050	0.0306	0.0000	0.1518
	$F_e(\beta)$	0.3643	0.2709	0.1675	0.0534	0.0000	0.1712
w/o CE	$P_l(\beta)$	0.5125	0.4396	0.2387	0.1399	0.0049	0.2671
	$F_e(\beta)$	0.4801	0.3538	0.2732	0.1508	0.0037	0.2523
w/o CAM	$P_l(\beta)$	0.7858	0.7687	0.7401	0.6879	0.5897	0.7144
	$F_e(\beta)$	0.7660	0.7421	0.7074	0.6523	0.5401	0.6815
MESD-CA	$P_l(\beta)$	0.8014	0.7887	0.7675	0.7267	0.6090	0.7386
	$F_e(\beta)$	0.7576	0.7388	0.7114	0.6668	0.5561	0.6861

from 0.63 to 0.00, a reduction of 0.63, and metric $F_e(\beta)$ falls from 0.50 to 0.00, a reduction of 0.50. These results indicate that GAN can only effectively predict VSM at lower thresholds, while its performance deteriorates markedly at higher thresholds, particularly at $\beta = 0.9$, where both $P_l(\beta)$ and $F_e(\beta)$ reach 0, rendering it nearly incapable of high-quality VSM prediction. Conversely, MESD-CA demonstrates remarkable stability, not only surpassing GAN at low thresholds but also maintaining excellent performance at high thresholds. These comparative experimental results validate the effectiveness and stability of the proposed MESD-CA in VSM decoding tasks, as evidenced by its significantly superior evaluation metrics and robust VSM prediction capability under high-threshold conditions, thereby demonstrating the algorithm's efficacy.

This study also conducted ablation experiments, with results presented in Table 1. The "Image" condition, where only single-modal visual image data were used as model input, exhibited notably low values for both $P_l(\beta)$ and $F_e(\beta)$, indicating that the model's decoding performance does not rely solely on visual images but rather results from the multimodal fusion of EEG and visual image data. When removing the IE (w/o IE), i.e., using only single-modality EEG signals as input, performance significantly declined across all thresholds. Particularly at high thresholds, $P_l(\beta)$ and $F_e(\beta)$ dropped to 0, with the mean $P_l(\beta)$ decreasing from 0.7386 to 0.1518 and $F_e(\beta)$ from 0.6861 to 0.1712. This suggests that single-modal EEG signals possess limited decoding capability at lower thresholds, but this capability diminishes as thresholds increase, highlighting the inadequacy of single-modality algorithms for high-standard VSM decoding tasks. When removing the CE (w/o CE), performance degradation was minimal at lower thresholds but substantial at higher thresholds; however, performance remained markedly superior to that of the model without IE. This is primarily because, after removing CE, the model retains both an EEG encoder and an image encoder, still enabling multimodal fusion. Nevertheless, due to the inability to decouple EEG signals, the feature extraction capacity of the EEG encoder is limited, resulting in reduced performance compared to MESD-CA. Upon removing the CAM (w/o CAM), $P_l(\beta)$ and $F_e(\beta)$ decreased by 0.0242 and 0.0046, respectively, indicating that CAM enhances the representation of channel signals to some extent by assigning higher attention weights to effective channels, thereby improving the overall model performance.

We calculated the evaluation metrics for different participants across various target object categories, with the experimental results presented in Table 2. Based on the mean evaluation metrics for different participants, Subject 2 outperforms the other two participants in both $P_l(\beta)$ and $F_e(\beta)$ metrics, with $P_l(\beta)$ and $F_e(\beta)$ values 0.0295 and 0.0317 higher than those of Subject 3, respectively. The primary reason for these inter-subject differences can be attributed to variations in individual brain structures and cognitive states, which result in distinct brain activity patterns. Similarly, analysis of the experimental results across different target object categories reveals that varying visual stimuli also contribute to differences in outcomes, primarily influenced by visual features such as the shape, color, and texture of the objects.

Table 2
Experimental results of different subjects in different categories.

Algorithm	Metric	Category				Average
		Projector	Fan	Power strip	Electric kettle	
Subject 1	$P_l(\beta)$	0.7972	0.7523	0.8196	0.7640	0.7832
	$F_e(\beta)$	0.7245	0.7087	0.7620	0.7442	0.7348
Subject 2	$P_l(\beta)$	0.8056	0.8221	0.7859	0.8313	0.8112
	$F_e(\beta)$	0.7459	0.7710	0.7599	0.7631	0.7599
Subject 3	$P_l(\beta)$	0.7456	0.7871	0.8001	0.7943	0.7817
	$F_e(\beta)$	0.7013	0.7455	0.7291	0.7369	0.7282
Average	$P_l(\beta)$	0.7828	0.7871	0.8018	0.7965	
	$F_e(\beta)$	0.7239	0.7455	0.7291	0.7369	

3.4. Brain topography analysis

This study introduces a CAM within the MESD-CA algorithm. This mechanism analyzes the significance of each channel in the original EEG signals for visual saliency decoding tasks, leading to the generation of corresponding channel attention weights. The channel attention weights correspond one to one with the EEG signal channels, totaling 59 dimensions, and are limited to positive values greater than zero. By multiplying the data from each channel by its corresponding weight [38], this mechanism aims to enhance the data from effective channels while suppressing the data from ineffective channels, ultimately obtaining the enhanced EEG signals to facilitate the subsequent extraction of features and the decoding of visual saliency.

We investigated the importance of different channels in visual focus decoding by analyzing the weights of channel attention. The experimental results are illustrated in Figs. 6 and 7, where we have plotted the distribution of channel attention weights on a brain topography to enhance visual clarity. In Fig. 6, the four topographical brain maps on the left correspond to different observational devices, helping to explore the impact of device categories on channel attention. The larger brain topography on the right represents the averaged result of the four maps on the left, aiming for a comprehensive analysis. From the right topographic map, it can be observed that channels Fp2, FC4, and PO7 exhibit relatively high values, followed by channels Oz, CP4, and F3, while other channels show relatively lower attention weights. According to current research on functional partitioning of the brain [6], vision-related regions include mainly the occipital and parietal lobes. The occipital lobe, as the primary visual cortex, is responsible for the initial perception of visual images from the retina to the brain. The parietal lobe functions as an advanced visual cortex, extracting useful information from visual images and integrating it with other modalities of information. PO7 is located at the junction of the occipital and parietal lobes, and its high channel attention weight suggests its importance in visual focus decoding. Another key electrode in the occipital region, Oz, is functional but has less impact. Furthermore, the frontal lobe also plays a crucial role in visual focus decoding [7], as reflected in the performance of the Fp2 and FC4 electrodes. By comparing the brain topography corresponding to different categories on the left side of Fig. 6, it is found that the channels Fp2 and Oz exhibit lower attention weights for the projector and fan than for the power strip and electric kettle, but higher weights in CP4. These findings suggest that the characteristics of the devices, such as category, color, and shape [39], can influence the operation of the visual attention mechanism.

This study further explores the impact of different visual focus positions on channel attention weights, as illustrated in Fig. 7. The top left corner provides a schematic diagram of visual field division, where the visual field is sectioned into five areas: center, top, bottom, left, and right. Set the radius R of the central area to 25% of the width. The channel attention weights were statistically analyzed when the visual focus was located within each area, followed by plotting the corresponding brain topography. The results generally resemble the distribution in Fig. 6, with Fp2, FC4, and PO7 remaining the most influential channels.

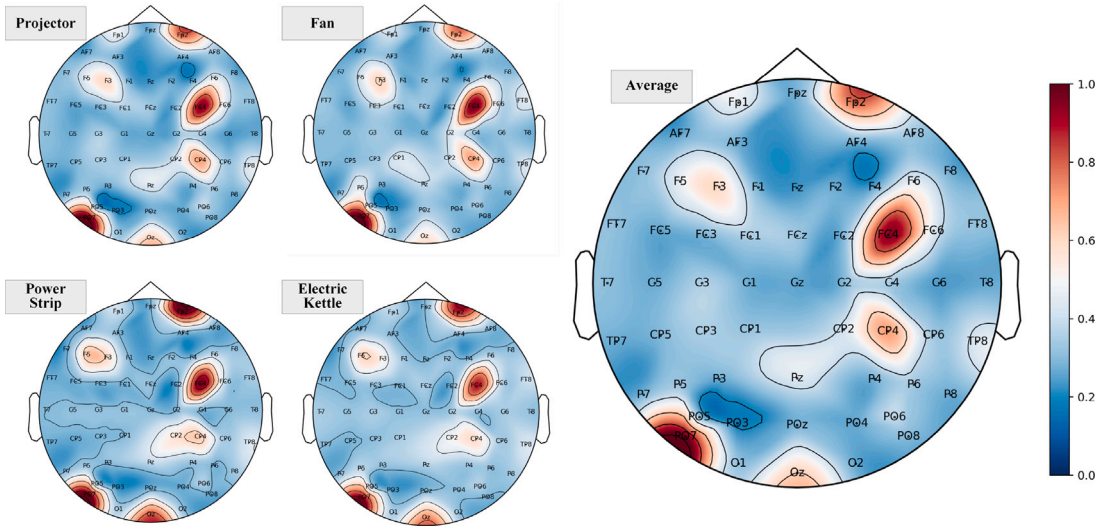

Fig. 6. Topographical brain maps corresponding to different classes.

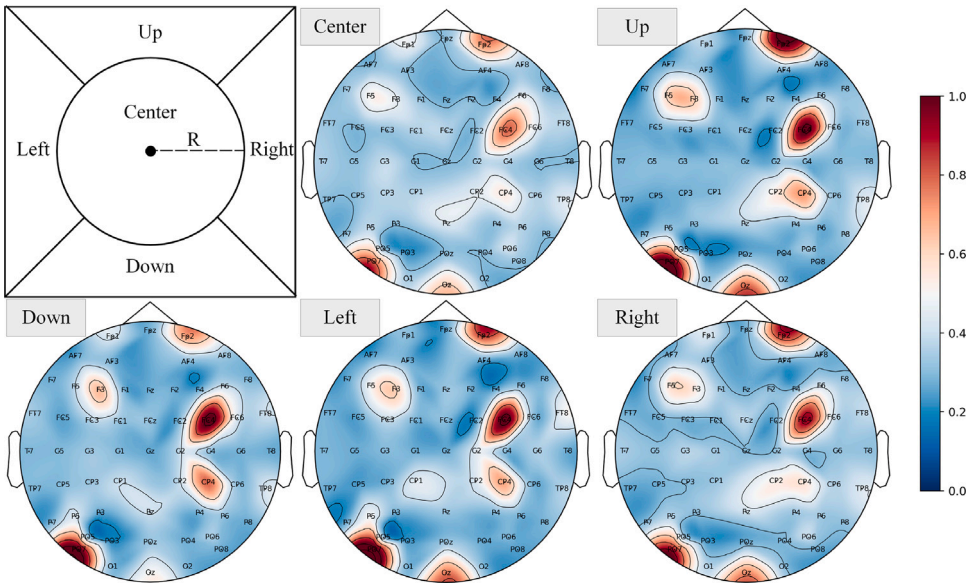

Fig. 7. Topographical brain maps corresponding to different visual focus positions.

Comparing brain topography across different regions reveals that the attention weights of key channels in the central area are significantly lower than those in the other four. This suggests that the central position serves as the default region for visual attention mechanisms, requiring less control when the visual focus is centered. In addition, the five topographic maps exhibit different states at certain key positions, indicating that different areas of the brain may functionally be involved in controlling visual attention, depending on the focus position. This suggests that while the overall visual attention mechanism operates similarly, each position demonstrates pattern differences.

4. Conclusions

This study proposes a novel method to predict visual saliency by decoding EEG signals. First, we designed a paradigm for visual saliency stimulation, where real objects were placed in front of participants to provide effective visual stimuli. This setup allowed us to collect a large volume of high-quality experimental data. Second, we proposed an improved VSM generation method based on a two-dimensional Gaussian distribution, which overcomes the structural complexity of traditional VSM generation approaches and reduces the difficulty of decoding VSM from EEG signals. Furthermore, we introduced a novel deep learning model, Multi-Encoder Single-Decoder with Channel Attention (MESD-CA). This model employs a two-stage training strategy to disentangle category and spatial information from EEG signals, integrating this information with visual image data for multimodal fusion, thereby improving VSM decoding performance. The GAN model was selected as a Baseline for comparison, and ablation experiments were conducted on the MESD-CA model. The analysis of the visualized VSM output indicates that MESD-CA demonstrates superior VSM decoding performance. In addition, the performance of the models was quantitatively evaluated using the PL and FE metrics. MESD-CA achieved PL and FE scores of 73.8% and 68.6%, respectively, outperforming the GAN model and other Baseline models. Compared to the MESD model without the CAM module, MESD-CA improved PL and FE scores by 2.4% and 0.5%, respectively, confirming the importance of the Channel Attention Mechanism in enhancing system performance. By combining the effects of various innovation points on model performance improvement, it is found that the algorithm's performance is most significantly enhanced after multimodal fusion, which is the core innovation point of the

model. Finally, we constructed brain topography maps based on the channel attention weights generated by CAM. The analysis revealed that the electrodes most critical for decoding VSM were Fp2, FC3, and PO7. In addition, different patterns of brain activity were observed when participants viewed different object categories or focused on objects at various locations in the visual field. In conclusion, the proposed MESD-CA model offers a novel and effective approach for decoding VSM from EEG signals, providing new insights into the neural mechanisms underlying visual saliency. Furthermore, this method has broad applicability and can be extended to other areas of EEG research, such as auditory and tactile studies, as well as intelligent brain control systems.

CRediT authorship contribution statement

Li Li: Conceptualization, Data curation, Formal analysis, Funding acquisition, Writing – original draft, Writing – review & editing. **Tongchao Lu:** Writing – review & editing, Writing – original draft, Visualization, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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